

SO(3) Equivariant Neural Networks

Lecture 1

What Should Rotate?

Symmetry, invariance, and equivariance from physical examples

90-minute journal-club lecture note

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Companion notebook included in the series package

Session map and learning goals

hook	symmetry	physics	break	AI translation	notebook	discussion
0			45			90 min

By the end of this lecture, you should be able to:

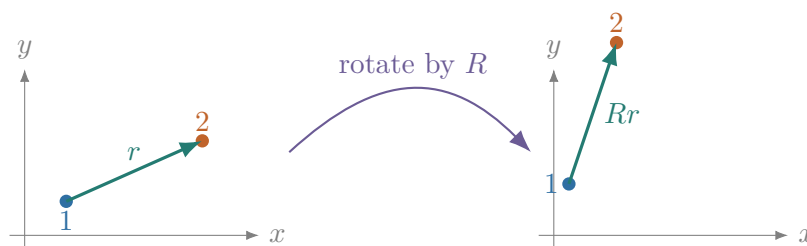
- recognize a three-dimensional rotation and name what it preserves;
- distinguish invariance from equivariance;
- write the expected transformation law for a scalar, vector, or rank-2 tensor;
- explain why data augmentation and architectural equivariance are not the same guarantee;
- test a candidate function numerically for rotational equivariance.

Key idea

The first design question for a physical neural network is not “Which architecture should we use?” It is “How should every input, hidden feature, and output transform when the experiment is rotated?”

1 Opening puzzle: rotate the entire experiment

Imagine two particles with relative displacement $r = r_2 - r_1$. A model predicts their interaction energy E and force F . We now rotate the particles and everything physical around them — the apparatus, any external fields — together by a rotation matrix R , while the coordinate axes stay fixed.



The numerical energy should stay the same, but the force should rotate:

$$E(Rr) = E(r), \quad F(Rr) = RF(r). \quad (1)$$

Both statements express symmetry, but they are different kinds of symmetry behavior.

Pause and discuss

Before reading further, classify these outputs: distance, velocity, pressure, electric field, stress, and total molecular energy. Which should stay unchanged, and which should rotate?

2 Rotations and the group SO(3)

A proper three-dimensional rotation is a real 3×3 matrix satisfying

$$R^\top R = I, \quad \det R = 1. \quad (2)$$

The first condition means that $R^{-1} = R^\top$, so rotations preserve dot products:

$$(Ru) \cdot (Rv) = u^\top R^\top R v = u \cdot v. \quad (3)$$

Consequently they preserve lengths and angles. The determinant condition removes reflections.

The word *group* means that rotations form a consistent transformation system:

- the identity I is a rotation;
- composing rotations gives a rotation;
- every rotation has an inverse;
- composition is associative.

We will rarely use these axioms explicitly. Their practical value is consistency: applying R_2 and then R_1 is equivalent to applying the single rotation $R_1 R_2$.

Math lens

$O(3)$ contains rotations and reflections. $SO(3)$ contains only proper rotations. This distinction matters for pseudovectors such as angular momentum and torque. Throughout the series, we test proper rotations only, so ordinary and axial vectors transform identically.

3 Invariance and equivariance

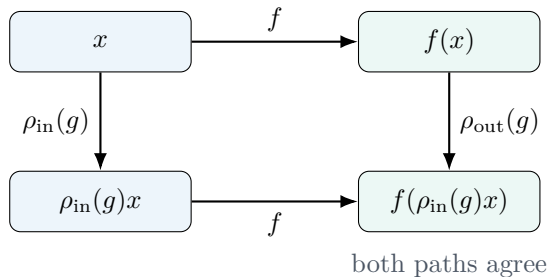
Let g transform an input x . A function is *invariant* when

$$f(gx) = f(x). \quad (4)$$

It is *equivariant* when the output transforms predictably:

$$f(\rho_{\text{in}}(g)x) = \rho_{\text{out}}(g)f(x). \quad (5)$$

Here ρ_{in} and ρ_{out} describe how the group acts on the input and output spaces. For vector inputs and outputs under rotations, both actions may simply be R .



An invariant is a special equivariant object for which $\rho_{\text{out}}(g) = 1$.

3.1 Three transformation types

Type	Components	Transformation	Examples
Scalar	1	$s' = s$	mass, energy, pressure
Vector	3	$v' = Rv$	position, velocity, force
Rank-2 tensor	3×3	$T' = RTR^\top$	stress, inertia, diffusion tensor

Physics lens

Physical laws are usually coordinate-independent, but that does not mean every quantity is invariant. The *law* is invariant in form; its vector and tensor quantities transform covariantly so that all observers obtain compatible predictions.

4 Worked example: dot and cross products

For $u, v \in \mathbb{R}^3$, the dot product is invariant:

$$(Ru) \cdot (Rv) = u \cdot v. \quad (6)$$

The cross product is equivariant under proper rotations:

$$(Ru) \times (Rv) = R(u \times v). \quad (7)$$

Therefore

$$y(u, v) = (u \cdot v)(u \times v) \quad (8)$$

is vector-equivariant: an invariant scalar rescales an equivariant vector.

This target will reappear in Lecture 5. It is simple enough to understand geometrically but requires a model to preserve vector information through a nonlinear computation.

5 What symmetry means for machine learning

Suppose a plain multilayer perceptron receives the six Cartesian components of (u, v) . Nothing in a dense layer prevents it from treating the x , y , and z axes differently.

AI lens

Three common strategies are:

1. **No symmetry treatment:** the model must discover rotational behavior from data.
2. **Data augmentation:** rotated examples encourage the desired behavior but do not guarantee it between training samples.
3. **Equivariant architecture:** every layer obeys the transformation law for all weights.

Architectural equivariance reduces the hypothesis space to functions compatible with the known physics.

Equivariance is not automatically better in every situation. It helps when the assumed symmetry is actually present, the transformation laws are correctly specified, and the excluded symmetry-breaking effects are not important for the task.

6 Notebook laboratory

Open `lecture_01_symmetry_and_equivariance.ipynb`. The notebook will:

1. generate proper random rotations;
2. verify Equation (2);
3. check the dot and cross products numerically;
4. define a reusable relative equivariance error;
5. compare an exactly equivariant function with an axis-dependent one.

The numerical residual should be near machine precision, typically 10^{-15} , for exact identities. A nonzero residual of order 10^{-2} or larger is not floating-point noise.

7 Questions and discussion

Question 1.1. For each quantity, state whether it is invariant, vector-equivariant, or rank-2 tensor-equivariant under proper rotations: energy, force, distance, velocity, and stress.

Question 1.2. Starting from $R^T R = I$, prove that the dot product is rotationally invariant.

Question 1.3. Why is predicting a vector a stronger test of an equivariant model than predicting only a scalar invariant?

Question 1.4. A model is trained with millions of randomly rotated examples. Does this prove that it is exactly equivariant? Explain.

Question 1.5. Why is $f(u, v) = (u \cdot v)(u \times v)$ equivariant? Identify the invariant and equivariant parts.

Question 1.6. Suppose an external laboratory field always points along the fixed z axis. Is the physical problem still fully SO(3) symmetric if the field is not rotated with the inputs?

Checkpoint

You are ready for Lecture 2 if you can complete the sentence: “An equivariant model does not require its output to stay fixed; it requires the output to _____.”